6.3 Trigonometric Ratios for Complementary Angles

Recall that two acute angles are said to be complementary if the sum of their measures is equal to $90°$.

What can we say about the acute angles of a right-angled triangle?

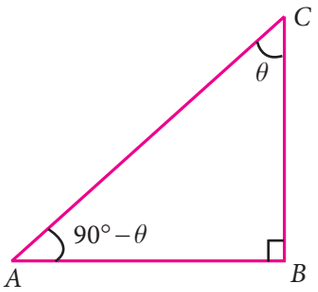


Fig. 6.13

In a right angled triangle the sum of the two acute angles is equal to 90°. So, the two acute angles of a right angled triangle are always complementary to each other.

In the above figure 6.13, the triangle is right-angled at B. Therefore, if $\angle C$ is θ, then $\angle A = 90° - \theta$.

We find that

$$\left.\begin{aligned} \sin\theta &= \frac{AB}{AC} \quad & \operatorname{cosec}\theta &= \frac{AC}{AB} \\ \cos\theta &= \frac{BC}{AC} \quad & \sec\theta &= \frac{AC}{BC} \\ \tan\theta &= \frac{AB}{BC} \quad & \cot\theta &= \frac{BC}{AB} \end{aligned}\right\} \ldots.(1)$$

Similarly for the angle (90ϒ– θ), We have

$$\left.\begin{aligned} \sin(90°-\theta) &= \frac{BC}{AC} \quad & \operatorname{cosec}(90°-\theta) &= \frac{AC}{BC} \\ \cos(90°-\theta) &= \frac{AB}{AC} \quad & \sec(90°-\theta) &= \frac{AC}{AB} \\ \tan(90°-\theta) &= \frac{BC}{AB} \quad & \cot(90°-\theta) &= \frac{AB}{BC} \end{aligned}\right\} \ldots..(2)$$

Comparing (1) and (2), we get

$$\begin{aligned} \sin\theta &= \cos(90°-\theta) \\ \cos\theta &= \sin(90°-\theta) \\ \tan\theta &= \cot(90°-\theta) \end{aligned} \qquad \left| \qquad \begin{aligned} \operatorname{cosec}\theta &= \sec(90°-\theta) \\ \sec\theta &= \operatorname{cosec}(90°-\theta) \\ \cot\theta &= \tan(90°-\theta) \end{aligned} \right.$$

Example 6.7 Express (i) $\sin 74°$ in terms of cosine (ii) $\tan 12°$ in terms of cotangent (iii) $\operatorname{cosec} 39°$ in terms of secant

Solution

(i) $\sin 74° = \sin(90° - 16°)$ (since, $90° - 16° = 74°$)

RHS is of the form $\sin(90° - \theta) = \cos\theta$

Therefore $\sin 74° = \cos 16°$

(ii) $\tan 12° = \tan(90° - 78°)$ (since, $12° = 90° - 78°$)

RHS is of the form $\tan(90° - \theta) = \cot\theta$

Therefore $\tan 12° = \cot 78°$

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